

# Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C<sub>60</sub> Graphullerene

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(Dated: June 19, 2026)

Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects need not add. In quasi-hexagonal C<sub>60</sub> graphullerene we quantify the omitted cross term with a synergy order parameter  $\mathcal{S}$  (meV/atom). PBE+DFT-D3 on tetramers and periodic  $n \times \text{C}_{60}$  supercells ( $n = 1\text{--}8$ ) gives opposite strain coefficients for N (−303 meV/%) and B (+57 meV/%) and  $|\mathcal{S}|$  up to  $\sim 32$  meV/atom at +3% strain, largest for P despite small  $|\alpha|$ . The pattern warns that sequential high-throughput screens can mis-rank stability when local dopant distortion couples nonlinearly to external strain.

## I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate  $E_{\text{tot}}(\epsilon, \delta)$  from *separate* doping-only and strain-only calculations [1, 2]. Synthetic carbon allotropes with extended  $\pi$  conjugation underpin optoelectronic applications, yet charge transport in molecular networks is often polaronic [3, 4]. Quasi-hexagonal (qHP) C<sub>60</sub> graphullerene [3, 4] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C<sub>60</sub> monolayers [3]. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 5, 6], but high-throughput pipelines often compose  $E_{\text{tot}}(\epsilon, \delta)$  from *separate* scans. Non-additive coupling is documented in other two-dimensional networks [1]; a quantitative *total-energy* cross term  $\mathcal{S}$  [Eq. (1)] has not been reported for substitutional B/N/P qHP C<sub>60</sub> tetramers and  $n = 1\text{--}8$  supercells under a single PBE+D3 protocol.

We address three questions on this model network: (1) how opposite linear strain coefficients  $\alpha$  emerge for donor vs. acceptor dopants; (2) how large the omitted cross term  $|\mathcal{S}|$  is when strain and substitution are applied jointly; and (3) whether  $|\mathcal{S}|$  tracks  $|\alpha|$  or instead reflects local structural distortion. Using  $\gtrsim 90$  converged CP2K tasks (Supplemental Material Table S2; see Supplemental Material at [URL will be inserted by publisher] for the full DFT inventory, extended PDOS, and transport-adjacent observables), we define  $\mathcal{S}$  on shared tetramer and periodic grids; Marcus  $\lambda$  and hopping observables (IPR/ $J$ ) are analyzed only in Supplemental Material Fig. S5–S6 and are not required for the non-additivity conclusions here. The central narrative anchor is that *local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy:  $|\mathcal{S}|$  can exceed what linear  $\alpha$  alone would suggest, so stability rankings from  $\epsilon = 0$  substitution

energies or from strain-only elastic increments need not survive a coupled load.

## II. COMPUTATIONAL METHODS

### A. Electronic-structure setup

All calculations use CP2K [7] with PBE [8] and Grimme DFT-D3 (Becke–Johnson damping) [9] (Supplemental Material for basis, SCF, and task inventory). We deliberately remain below the Koopmans/rVV10 level of Ref. 3 to obtain a uniform, size-converged  $(\epsilon, \delta)$  scan rather than absolute polaron binding energies.

### B. Notation and conventions

Biaxial strain is denoted  $\epsilon$  (percent); dopant species  $\delta \in \{B, N, P\}$ . The synergy order parameter is always  $\mathcal{S}$  (calligraphic) in meV/atom [Eq. (1)].  $\pi$ -projected densities of states are written  $\pi$ -DOS (never bare “pi” in equations).

**Rigid-strain protocol.** Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported  $\alpha$  and  $\mathcal{S}$  under this protocol are *upper bounds* on coupling under load until the ionic-relaxation benchmark in Table S3 converges.

**Plane-wave cutoff (size scaling).** Archived CP2K inputs use 400 Ry for  $n \leq 4$  and 350 Ry for  $n \geq 6$  (Table S2); comparisons at fixed  $n$  always share the same cutoff. Cross- $n$  trends therefore mix a modest basis change at  $n = 6$ ; we do *not* interpret the reported N  $\mathcal{S}$  sign change between  $n = 6$  and  $n = 8$  as a physical size effect until a matched 400 Ry  $n = 6$  control completes (input `size_6x60_N_pos3pct_cutoff400`).

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### C. Substitutional doping and grids

Tetramer models ( $4 \times C_{60}$ , 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates, random seed 42), not a thermodynamic minimum. A second seed 137 tetramer ensemble (Table S4) tests whether  $\alpha$  and tetramer-scale  $\mathcal{S}$  are sensitive to dopant placement; periodic  $\mathcal{S}(n)$  claims remain on the seed 42 periodic protocol until seed 137 periodic points are added. Periodic scaling cells place one dopant per  $C_{60}$  ( $n = 1\text{--}8$ ), so nominal concentration falls from  $\sim 5\%$  to  $\sim 1.7\%$  per atom at  $n = 4$ . Tetramer  $\alpha$ , gaps, and Supplemental Material IPR/ $J$  trends are therefore not directly interchangeable with periodic  $\mathcal{S}(n)$  without a matched-concentration benchmark; all  $\mathcal{S}$  claims use the periodic protocol. In total,  $\gtrsim 90$  converged neutral tasks support the quantitative claims below (Table S2). Charged polaron geometry optimizations and vertical single-points for Marcus reorganization energies follow the same functional and basis choices but are reported separately in Supplemental Material Fig. S5 once converged; they do not enter the  $\mathcal{S}$  definition in Eq. (1).

### D. Energy observables

Linear strain sensitivity  $\alpha = dE/d\epsilon$  (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table S1). Positive  $\alpha$  means tension *raises* total energy; negative  $\alpha$  means tension *lowers* it under the rigid-strain protocol. Substitution energy  $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$  is reported relative to pristine tetramers at the same strain (no chemical-potential reference). In sequential high-throughput screens one often estimates  $E_{\text{tot}}(\epsilon, \delta) \approx E_f(\epsilon=0) + E_{\text{strain}}(\epsilon)$ ; the neglected term is precisely  $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows, the magnitude of this error is best gauged against the substitution-energy scale: at  $n = 4$  and +3% strain, P has  $|\mathcal{S}| \approx 24$  meV/atom while the per-atom substitution scale is  $|E_f| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$  eV/atom ( $|E_f| \approx 16$  eV per P site,  $n_{\text{dop}}=12$ , Table S1)—i.e.  $|\mathcal{S}|/|E_f|_{\text{per atom}} \sim 3\%$ , large enough to reorder stability margins in sequential doping-then-strain screens even though  $|\mathcal{S}|$  exceeds the rigid per-atom strain increment ( $\alpha \times 3\%/N_{\text{atoms}} \lesssim 0.3$  meV/atom for P).

**Non-additive coupling metric.** We quantify strain–doping cross terms with the synergy order parameter  $\mathcal{S}$ :

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where  $\epsilon$  is biaxial strain,  $\delta \in \{B, N, P\}$ , and  $\Delta E(\epsilon, \delta)$  is the energy per atom relative to the pristine, zero-strain reference at the same supercell size  $n \times C_{60}$ . The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives

$\mathcal{S} = 0$ ;  $\mathcal{S} < 0$  indicates cooperative stabilization beyond the two-scan sum, whereas  $\mathcal{S} > 0$  indicates cooperative destabilization.

To first order in strain,  $\mathcal{S}(\epsilon, \delta) \approx (\alpha_\delta - \alpha_0)\epsilon$  when  $\alpha_\delta = dE/d\epsilon$  is evaluated for the doped cell and  $\alpha_0$  for the pristine reference at the same  $n$  (rigid-strain protocol); this links the mechanical coefficient  $\alpha$  to the cross term at small  $\epsilon$ .

Periodic-cell  $\mathcal{S}$  values are obtained from archived total energies via post-processing of converged CP2K outputs (Supplemental Material Table S2). We report  $\mathcal{S}$  for total energy only; an analogous gap cross term  $\mathcal{S}_{\text{gap}}$  is not computed here—main-text panel (a) shows that gap slopes need not track  $|\mathcal{S}|$ .

**Localization and coupling (Supplemental Material Fig. S6).** Following Capobianco *et al.* [3], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling  $J$  from tetramer dimer fragments. IPR follows Supporting Information Eq. (S8),  $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ , on a Mulliken-projected orbital grid;  $J$  is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant  $J \simeq 75\text{--}81$  meV reported for pristine qHP networks [3], but they provide an internally consistent trend under coupled  $(\epsilon, \delta)$ .

**Polaron binding (Supplemental Material).** Marcus reorganization energies follow the vertical–adiabatic decomposition [3]:  $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ , with separate internal ( $\lambda_{\text{loc}}^\pm$ ) and environmental ( $\lambda_{\text{env}}^\pm$ ) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Supplemental Material Fig. S5.

### E. Validation benchmarks (Supplemental Material)

**(i) Ionic relaxation (Table S3).** Four fixed-cell GEO.OPT corners (pristine and P-doped at  $\epsilon = 0, +3\%$ ) quantify how much rigid  $\mathcal{S}$  overestimates the cross term when atoms relax under the same biaxial cell. **(ii) Alternate dopant ensemble (Table S4).** Tetramer B/N/P at seed 137 repeats the  $\epsilon = 0, +3\%$  rigid grid to bound placement sensitivity. **(iii) Cutoff control (Table S2).** A single  $6 \times C_{60}$  N @ +3% task at 400 Ry isolates basis-set confounding from electrostatic size trends.

## III. RESULTS

We organize the results around total-energy non-additivity: Sec. III A summarizes band-edge context; Sec. III B reports linear strain coefficients and  $E_f$ ; Sec. III C gives  $\mathcal{S}(n)$  at +3% strain across dopants and supercell sizes (IPR/ $J$  and Marcus  $\lambda$  remain in Supplemental Material Fig. S5–S6).

## A. Electronic structure under strain and doping

Figure 1(a) reports HOMO–LUMO gaps from tetramer single-points at  $\epsilon \in \{-5, 0, +5\}\%$ . B (acceptor) introduces states near the valence manifold and compresses the  $\pi$  gap toward zero at  $\epsilon = 0$ ; N (donor) pushes electron density into the conduction manifold; P perturbs the network more heterogeneously because of its larger covalent radius and longer cage bonds. Tensile strain generally narrows the pristine gap while dopant-specific slopes differ: B gaps remain the smallest across the grid, whereas N and P retain finite gaps at  $\epsilon = 0$  (panel (a)). Figure 1(b) shows representative  $\pi$ -DOS for B @  $\epsilon = 0$  (valence-edge and conduction-edge coloring follows Ref. 3). These band-edge trends contextualize the mechanical and non-additive energy results but are not used to infer  $\mathcal{S}$ .

## B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ( $|\alpha| \approx 1$  meV/%). Substitution opens an order-of-magnitude spread in  $\alpha$ : N ( $-303.2$  meV/%, tension stabilizes the doped cell), B ( $+56.9$  meV/%, tension destabilizes), and P ( $-19.1$  meV/%, weakly stabilizing) (Table S1). The opposite sign for N vs. B persists across the  $\epsilon \in \{-5, \dots, +5\}\%$  grid and is not an artifact of a single strain point. At  $\epsilon = +5\%$ , N remains the most tension-stabilized dopant ( $\alpha$  most negative), whereas B shows the strongest destabilization under tension; P sits between these extremes with a modest negative  $\alpha$ , foreshadowing why its linear coefficient poorly predicts the large  $|\mathcal{S}|$  at  $+3\%$  in periodic cells. Substitution energies at  $\epsilon = 0$  span  $E_f = +73.4$  eV/dopant (B),  $-115.3$  eV/dopant (N), and  $-15.8$  eV/dopant (P), with  $n_{\text{dop}} = 12$ – $13$  counted from input coordinates (Table S1). Figure 1(c) highlights the  $\alpha$  split; ranking dopants by  $\epsilon = 0$   $|E_f|$  alone is misleading once  $+3\%$  strain is applied.

## C. Non-additive strain–doping coupling

Figure 1(d) tracks  $\mathcal{S}(n)$  with one dopant per  $\text{C}_{60}$  at  $+3\%$  biaxial strain. At  $n = 4$ ,  $\mathcal{S} = -6.2$  meV/atom (B),  $+3.0$  meV/atom (N), and  $-23.7$  meV/atom (P); P shows the largest magnitude despite the smallest  $|\alpha|$  on the tetramer grid (Table S1). Across fifteen coupled points ( $n \in \{1, 2, 4, 6, 8\}$ , B/N/P),  $|\mathcal{S}|$  reaches  $\sim 32$  meV/atom for P at  $n = 1$  and remains  $\mathcal{O}(10)$  meV/atom at larger  $n$  for B and P, whereas N is smaller in magnitude at  $n = 4$  but comparable at  $n = 1$ . For N,  $\mathcal{S}$  changes sign between  $n = 6$  ( $+2.4$  meV/atom) and  $n = 8$  ( $-2.3$  meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until a matched 400 Ry  $n = 6$  control and Table S3–S4 benchmarks complete (Sec. II E). Size trends for B and P at fixed cutoff are more monotonic:  $|\mathcal{S}|$  decreases from the  $n = 1$  corner toward  $n = 4$

but remains  $\mathcal{O}(10)$  meV/atom at  $n = 4$ – $8$ . We report provisional  $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$  fits in Supplemental Material but withhold asymptotic  $\mathcal{S}_\infty$  claims until cutoffs are matched across  $n$ . Negative  $\mathcal{S}$  (B, P at  $n = 4$ ) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive  $\mathcal{S}$  (N at  $n = 4$ ) means sequential screens *overestimate* stability under load.

## IV. DISCUSSION

**Context.** Capobianco *et al.* [3] established qHP  $\text{C}_{60}$  as a polaronic network whose mobility gain over vdW  $\text{C}_{60}$  tracks enhanced interfullerene coupling rather than reduced reorganization energy. Li *et al.* [2] and Khan and Leuenberger [5] show that strain and impurity engineering independently move band edges in fullerene monolayers, while Materials2024untangling reports non-additive strain–doping effects in other 2D networks. Our contribution is complementary: at fixed PBE+D3, we quantify the *total-energy* cross term  $\mathcal{S}$  omitted when those knobs are applied sequentially, distinct from hopping metrics analyzed only in Supplemental Material Fig. S6. Where Capobianco *et al.* [3] emphasize that mobility gains track interfullerene  $J$  rather than reduced  $\lambda$ , our results show that *total-energy* stability under simultaneous strain and substitution is likewise not reducible to a sum of one-parameter scans—a caution that applies to materials discovery pipelines even when transport figures of merit are not computed.

### Mechanistic synthesis (three coupling classes).

**(i) Opposite  $\alpha$  for N vs. B.** N introduces excess  $\pi$  electrons that strengthen bonding under tension (negative  $\alpha$ ), whereas B creates electron deficiency and antibonding  $\pi^*$  character that rises with lattice expansion (positive  $\alpha$ )—consistent with donor vs. acceptor  $\pi$ -counting and the PDOS shifts in Fig. 1(a,b). **(ii) Large  $|\mathcal{S}|$  for P despite small  $|\alpha|$ .** P’s covalent radius ( $\sim 107$  pm vs.  $\sim 77$  pm for C) and longer P–C bonds impose intrinsic cage distortion before external load is applied; tensile strain then couples to this pre-existing local stress field rather than to a pristine cage, yielding a cross term that is invisible in separate doping-only and strain-only scans but dominant when both are engaged—consistent with P’s intermediate  $E_f$  yet largest  $|\mathcal{S}|$  at  $+3\%$  and with the weak tetramer  $|\alpha|$ . **(iii) N  $\mathcal{S}$  sign change with  $n$ .** With one dopant per  $\text{C}_{60}$ , electrostatic fields dilute at large  $n$ ; any N sign flip between  $n = 6$  and  $n = 8$  is confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. II E). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of  $\alpha$ , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large  $n$ ).

**(iv) Linear  $\mathcal{S}$ – $\alpha$  check.** Using tetramer  $\alpha$  (Table S1) in  $\mathcal{S} \approx (\alpha_\delta - \alpha_0)\epsilon$  at  $\epsilon = +3\%$  and dividing by 240 sites gives  $\sim +0.7$ ,  $-3.8$ , and  $-0.2$  meV/atom for B/N/P—far smaller than the periodic  $n=4$  values ( $-6.2$ ,  $+3.0$ ,

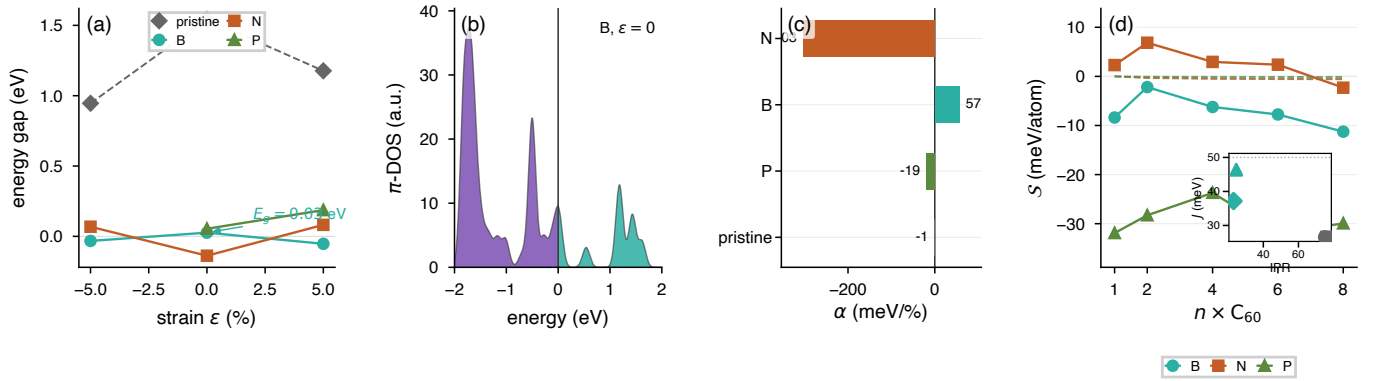


FIG. 1. Coupled  $(\epsilon, \delta)$  response in doped qHP  $C_{60}$  (PBE+D3). (a) HOMO-LUMO gap vs. strain. (b)  $\pi$ -DOS for B @  $\epsilon=0$  (Electron SI palette). (c) Mechanical  $\alpha$ . (d)  $S(n)$  at +3%; inset: schematic IPR/ $J$  grid (details in Supplemental Material Fig. S6).

−23.7 meV/atom). The gap shows that tetramer-scale linear mechanics and 5% doping do not quantitatively predict periodic  $S$ ; for P, nonlinear coupling dominates beyond first-order  $(\alpha_\delta - \alpha_0)\epsilon$ .

**(v) Significance versus DFT uncertainty.** At  $n=4$  and +3% strain,  $|S| \sim 24$  meV/atom is  $\sim 3\%$  of the per-atom substitution-energy scale (Table S1) but still  $\gg$  typical rigid per-atom strain increments from  $\alpha$ . PBE formation energies for molecular solids can carry  $\mathcal{O}(10^2)$  meV/atom uncertainties; we therefore treat  $S$  as a *coupled-scan correction* to sequential workflows rather than an absolute formation-energy prediction, and flag cases where  $|S|$  exceeds the rigid strain increment as the primary screening hazard.

**Design implications.** At +3% strain, P carries the largest additive-approximation error in  $S$  ( $|S| \sim 24$ –32 meV/atom); co-design should treat P under tension as a *coupled* stability problem, not  $|E_f|$  at  $\epsilon = 0$  alone. Where  $S < 0$  (B and P at  $n=4$ ), coupled tension can stabilize doped cells beyond additive predictions; where  $S > 0$  (N at  $n=4$ ), sequential screens overestimate stability under load. For example, a workflow that ranks P vs. B stability at  $\epsilon = 0$  from  $E_f$  alone and then applies +3% strain sequentially would miss a  $\sim 24$  meV/atom correction for P—comparable to several percent of the substitution-energy scale and much larger than the rigid strain increment from  $\alpha$ . Gap closing [Fig. 1(a)] does not track  $|S|$ ; we do not compute  $S_{\text{gap}}$ , so photonic or transport targets cannot be inferred from total-energy  $S$  alone. For B at  $n = 4$ , both negative  $S$  and positive  $\alpha$  imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating  $S$ .  $N$   $S$  vs.  $n$  is reported for completeness but excluded from design rules until Table S2–S4 controls converge.

**Limitations and outlook.** Three protocol limits are actively benchmarked (Sec. II E): rigid strain (Table S3), fixed-seed ensembles (Table S4), and mixed 400/350 Ry cutoffs across  $n$  (Table S2 control task). Additional

limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10;  $S_{\text{gap}}$  not quantified; Marcus  $\lambda$  pending charged vertical SP (Supplemental Material Fig. S5).

## V. CONCLUSIONS

In discrete covalent molecular networks, nonlinear coupling between dopant-induced local distortion and external strain can dominate sequential strain-doping energy estimates. Using qHP  $C_{60}$  as a model system, we quantified this cross term with  $S$  (meV/atom):  $|S|$  is not simply tied to linear  $\alpha$  (P shows the largest magnitude despite small  $|\alpha|$ ), and additive scans can mis-rank stability at the few-percent level of the substitution-energy scale. Opposite N vs. B strain coefficients and  $|S|$  at  $n = 4$  for B and P show that donor/acceptor chemistry and load must be co-designed; size-dependent N trends await cutoff and placement controls (Tables S2–S4). Ionic-relaxation, alternate-seed, and matched-cutoff benchmarks remain to test generality beyond the rigid seed 42 protocol applied here.

## ACKNOWLEDGMENTS

We acknowledge computational resources and support from Xiamen University.

## DATA AVAILABILITY

The data that support the findings of this study are openly available in the `sci-simukit` repository at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged CP2K inputs and outputs under `dft_results/`, analysis JSON under `experiments/analysis/`, and figure-generation scripts under `paper/figures/`. A

tagged release matching the manuscript revision will be issued upon acceptance.

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- [1] S. Chen, H. Liu, and K. Wang, *Advanced Materials* **36**, 2301234 (2024).
- [2] H. Li, M. A. Shakoory, and R. Wang, *Journal of Applied Physics* **136**, 015101 (2024).
- [3] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, *Nano Letters* **24**, 7891 (2024).
- [4] Y. Yang, W. Zhang, H. Liu, and K. Li, *Applied Surface Science* **536**, 147815 (2021).
- [5] M. A. Khan and M. N. Leuenberger, *Physical Review Materials* **9**, 034001 (2025).
- [6] B. Peng and M. Pizzochero, *Chemical Communications* **61**, 1234 (2025).
- [7] J. VandeVondele, M. Krack, F. Mohamed, M. Parrinello, T. Chassaing, and J. Hutter, *Computer Physics Communications* **167**, 103 (2005).
- [8] J. P. Perdew, K. Burke, and M. Ernzerhof, *Physical Review Letters* **77**, 3865 (1996).
- [9] S. Grimme, S. Ehrlich, and L. Goerigk, *Journal of Computational Chemistry* **32**, 1456 (2011).